

FETAL GROWTH RESTRICTION

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Fetal growth restrictions(FGR)

- **There are wide variety of reasons why a baby may be born small including congenital anomalies , fetal infections and chromosomal abnormalities .
- **however, most babies that are born small are either **constitutionally small** (i.e healthy but born to small parents and fulfilling their genetic growth potential) or are small secondary to abnormal placental function and have **FGR**.
- **FGR is a major cause of neonatal and infant morbidity and mortality.

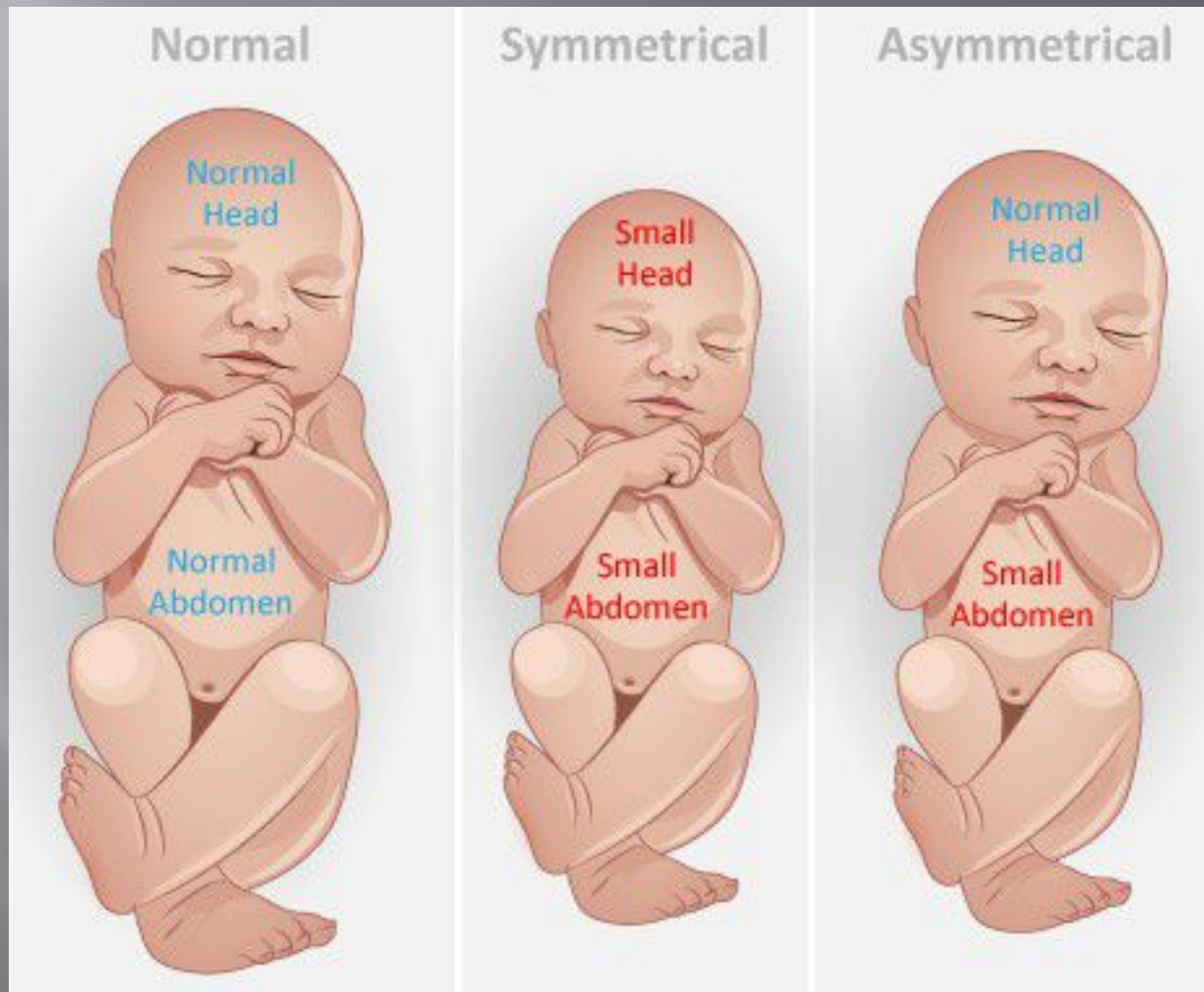
***In addition there is an increasing body of evidence that certain adult disease (such as diabetes and hypertension) are most common in adult who were born with FGR.

Definition and incidence :

** **FGR is defined as a failure of a fetus to achieve its genetic growth potential.** This usually results in a fetus that is small for gestational age (SGA).

**SGA means that the weight of the fetus is less than the tenth centile for its gestation. Other cut – off points (e.g the third centile) can be used.

**The term SGA and FGR are not synonymous.



What factors affect fetal weight?

1. Sex

term males 150 gm
heavier and 0.9 cm
longer than females

2. Parity

1st born infants smaller
effect loss after 3rd
birth

3. Race, ethnicity, nationality

4. Altitude

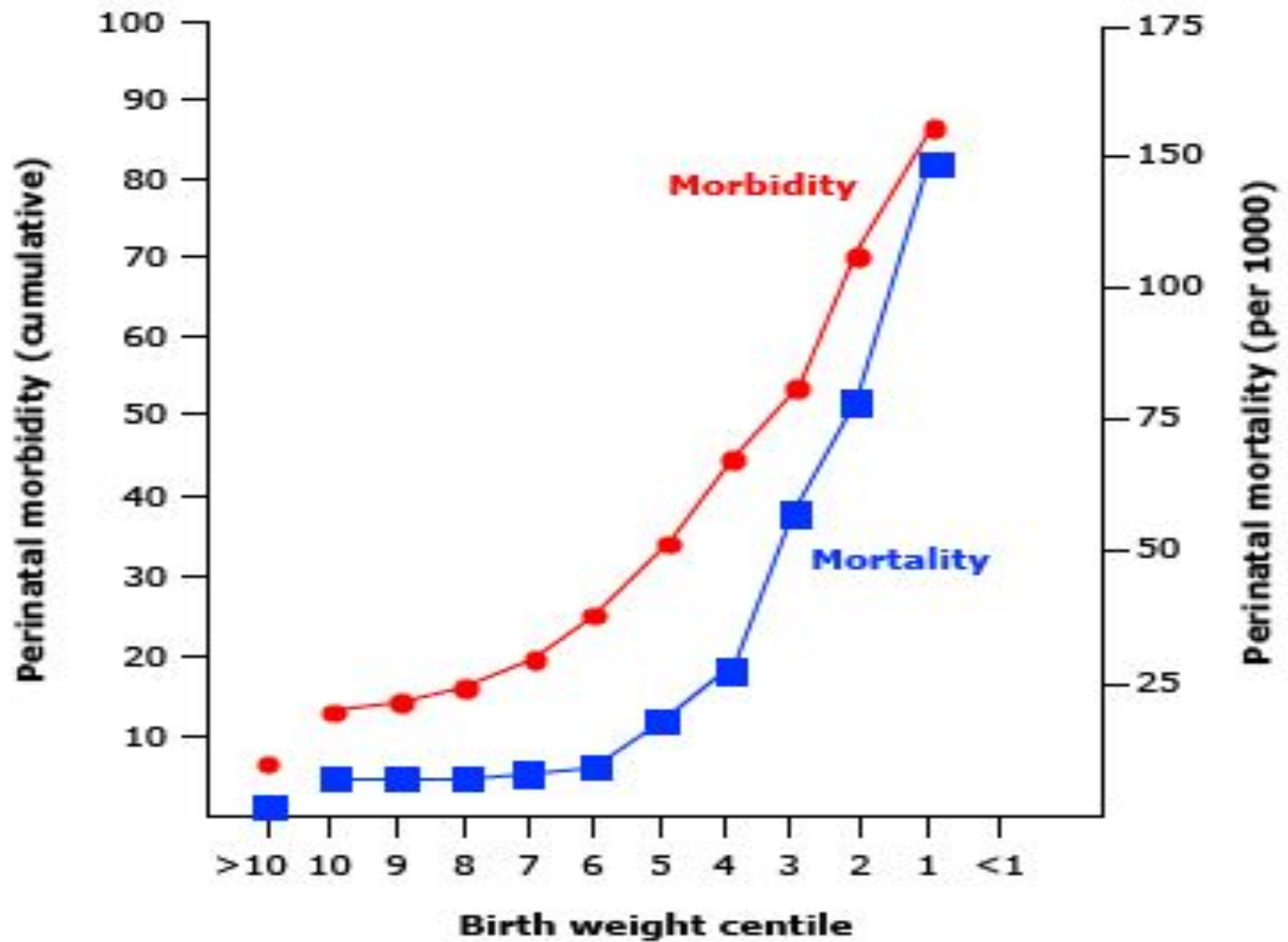
Denver population growth
curves under estimate
weights of infants born
at sea level

5. Maternal size

maternal pre-pregnancy
weight and pregnancy
weight gain correlate
with fetus size

Normal Intrauterine Growth

Stage 1	Stage 2	Stage 3
Hyperplasia	Hyperplasia/ hypertrophy	Hypertrophy
weeks 4-20	weeks 20-28	weeks 28-40
Rapid mitosis	Declining mitosis	Rapid hypertrophy
Increasing DNA content	Increasing cell size	Rapid increasing cell size
		rapid accumulation of fat, muscle, connective tissue
Symmetric	Mixed- asymmetric	Asymmetric

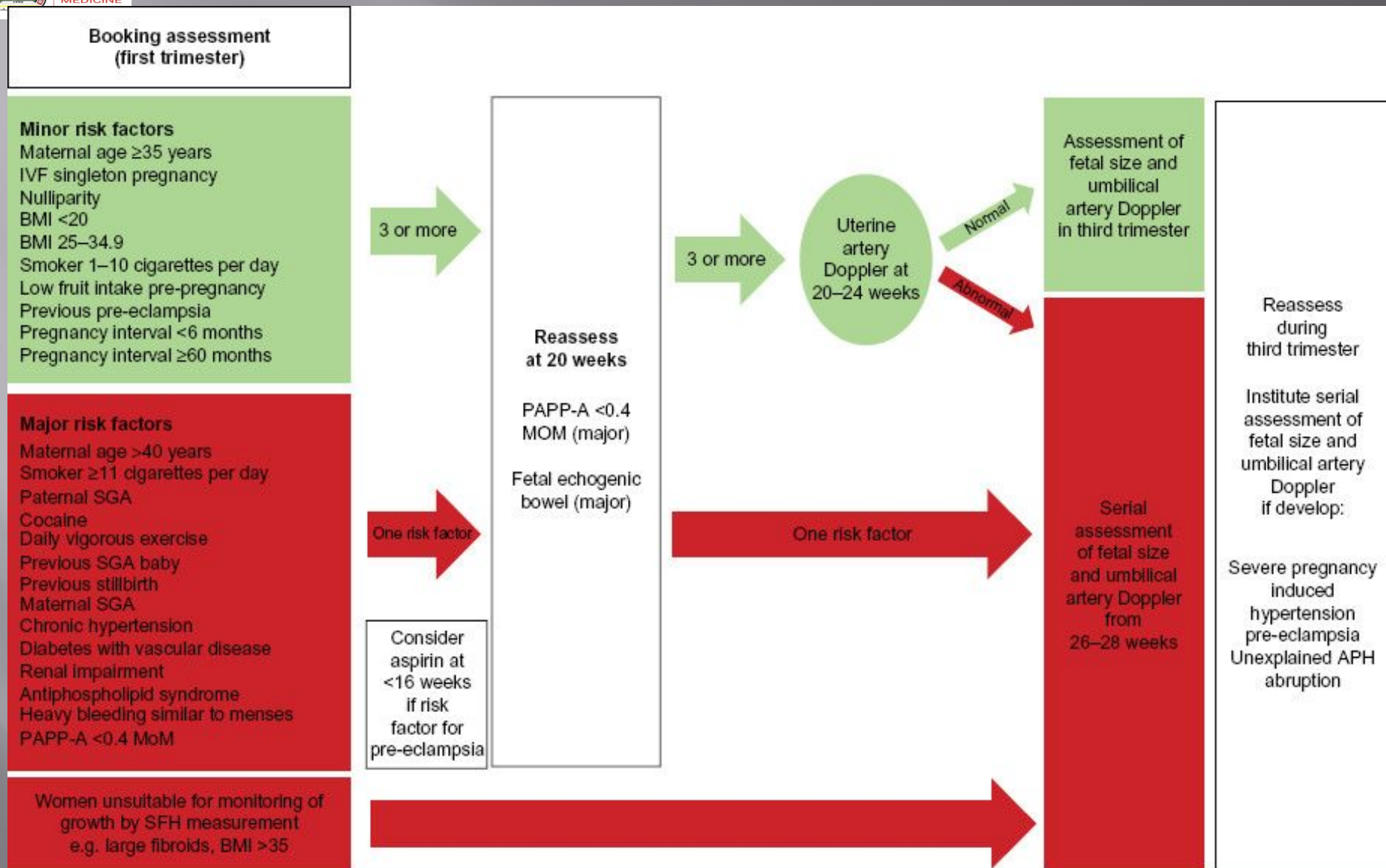


What are the risk factors for a SGA fetus/neonate? What is the optimum method of screening for the SGA fetus/neonate and care of “at risk” pregnancies?

All women should be assessed at booking for risk factors for a SGA fetus/neonate to identify those who require increased surveillance.

Women who have a major risk factor should be referred for serial ultrasound measurement of fetal size and assessment of wellbeing with umbilical artery Doppler from 26–28 weeks of pregnancy .

Women who have three or more minor risk factors should be referred for uterine artery Doppler at 20–24 weeks of gestation



Risk assessment must always be individualised (taking into account previous medical and obstetric history and current pregnancy history). Disease progression or institution of medical therapies may increase an individual's risk.

They are grouped into 2 main categories :

1. Reduced fetal growth potential (directly affect the intrinsic growth potential of the fetus) .

- a. Aneuploidies, e.g. trisomy 18,
- b. Single gene defects e.g. Seckel's syndrome.
- c. Structural abnormalities, e.g. renal agenesis.
- d. Intrauterine infection, Cytomegalovirus(CMV), Toxoplasmosis

2. Reduced fetal growth support:

a. Maternal(systemic) factors:

- **Under-nutrition (globally the major cause of FGR) , e.g. poverty, eating disorders.**
- **Maternal hypoxia, e.g. living at altitude, cyanotic heart disease.**
- **Drugs, e.g. alcohol, cigarettes, cocaine.**

Smoking, will increase the amount of carboxyhaemoglobin in the maternal circulation, effectively reduces the amount of oxygen available to the fetus, thus causing FGR.

alcohol and cocaine, probably act through multiple mechanisms affecting fetal enzyme systems, placental blood flow and maternal substrate levels.

b. Placental factors:

*****Reduced uteroplacental perfusion**, e.g. inadequate trophoblast invasion, sickle cell disease, multiple gestation.

*****Reduced fetoplacental perfusion**, e.g. single umbilical artery, twin-twin transfusion syndrome.

In developed countries, the most common cause of FGR is **poor placental function secondary to inadequate trophoblast invasion of the spiral arteries**. This results in **reduced perfusion** of the intracotyledon space which in turn leads to abnormal development of the terminal villi and impaired transfer of oxygen and nutrients to the fetus.

Less frequently, **reduced perfusion** can occur from other conditions such as **maternal sickle cell disease and the antiphospholipid syndrome**. **Multiple pregnancy** usually results in a sharing of the uterine vascularity, which cause a relative reduction in the blood flow to each placenta.

Pathophysiology :

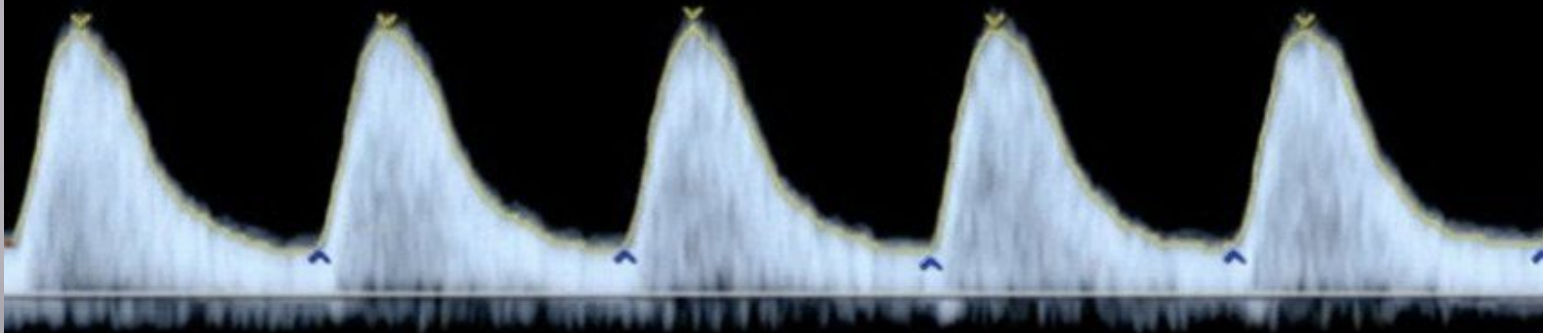
FGR is frequently classified as

1. Symmetrical small fetuses are normally associated with factors that directly impair fetal growth. Such as chromosomal disorders and fetal infections.

2. Asymmetrical growth restriction is classically associated with uteroplacental insufficiency which leads to reduced oxygen transfer to the fetus and impaired excretion of CO₂ by the placenta.

****A fall in PO₂ and a rise in *p*CO₂ in the fetal blood induces a chemoreceptors response in the fetal carotid bodies with resulting vasodilatation in the fetal brain, myocardium and adrenal glands and vasoconstriction in the kidneys, splanchnic vessels, limbs and subcutaneous tissues.**

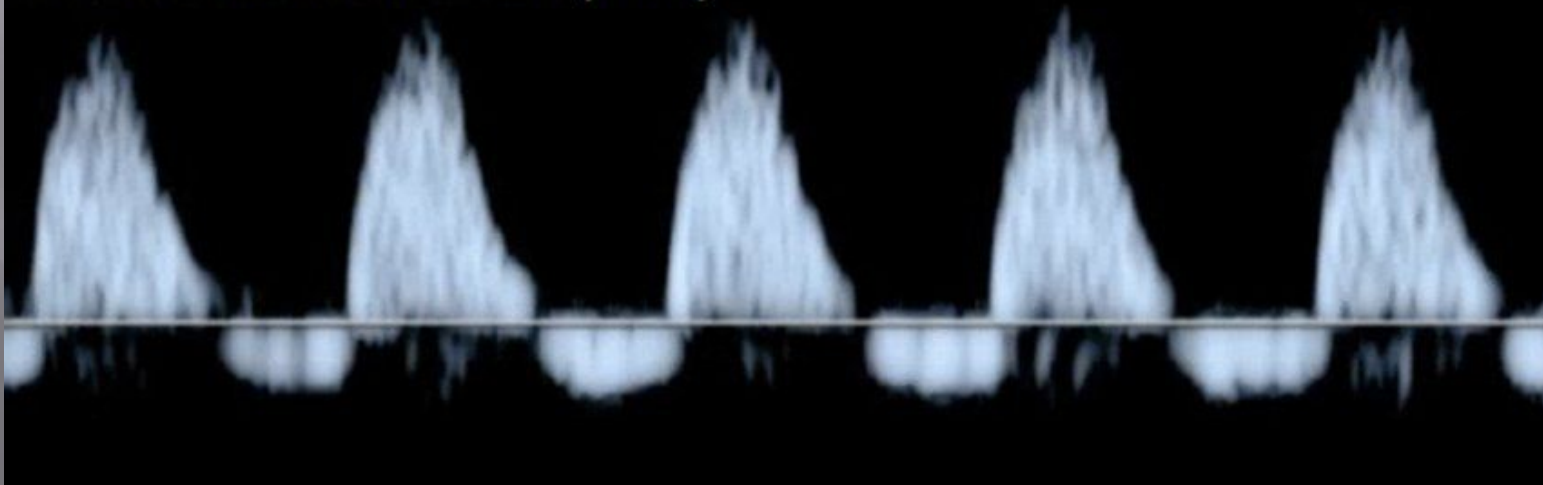
Decreased end diastolic flow



Absent end diastolic flow (AEDF)



Reversed end diastolic flow (REDF)



- **The liver circulation is also severely reduced.** Normally, 50% of the well oxygenated blood in the umbilical vein passes to the right atrium through the ductus venosus, eventually to reach the fetal brain. With the reminder going to the portal circulation in the liver.
- **When there is fetal hyoxia, more of the well-oxygenated blood from the umbilical vein is diverted through the ductus venosus, which means that the liver receives less.**
- **The result of all these circulatory changes is an asymmetrical fetus with relative brain sparing, reduced abdominal girth and skin thickness.** The vasoconstriction in the fetal kidneys results in impaired urine production and oligohydramnios. The fetal hypoxaemia also leads to severe metabolic changes in the fetus reflecting intrauterine starvation.

- **Antenatal fetal blood sampling has shown **reduced levels of nutrients such as glucose and amino acids** (especially essential amino acids) and of **hormones such as thyroxine and insulin**. There are **increased levels of corticosteroids and catecholamines**, which reflect the increased perfusion of the adrenal gland.

- **Haematological changes also reflects the chronic hypoxia, with **increased levels of erythropoietin and nucleated red blood cells**.

- **Chronic fetal hypoxia in FGR may eventually lead to fetal **acidemia, both respiratory and metabolic** which if prolonged can lead to intrauterine death if the fetus is not removed from its hostile environment.

*** Thereafter, the most precise way of assessing fetal growth is by **ultrasound biometry** (biparietal diameter, head circumference, abdominal circumference and femur length) **serially at set time intervals (usually of 4 weeks and no less than 2 weeks).**



- ** When a diagnosis of SGA has been made, the next step is to clarify whether the baby is **normal and simply constitutionally small or whether it is FGR.**

- ** A comprehensive ultrasound examination of the fetal anatomy should be made looking for fetal abnormalities that may explain the size. **Even if the anatomy appears normal, the presence of a normal amniotic fluid volume raises the suspicion of a fetal genetic defect and the parents should be counselled accordingly.** Amniocentesis and rapid fetal karyotype should be offered.

- ** Features suspicious of uteroplacental insufficiency are an **asymmetrically growth restricted fetus with a relatively small abdominal circumference, oligohydramnios and a high umbilical artery resistance.**

Optimal method and frequency of fetal surveillance in SGA In a high-risk population

the use of umbilical artery Doppler has been shown to reduce perinatal morbidity and mortality.

Umbilical artery Doppler should be the **primary surveillance** tool in the SGA fetus. When umbilical artery Doppler flow indices are **normal** it is reasonable to repeat surveillance **every 14** days.

More frequent Doppler surveillance may be appropriate in a severely SGA fetus When umbilical artery Doppler flow indices are **abnormal** (pulsatility or resistance index $> +2$ SDs above mean for gestational age) and delivery is not indicated repeat surveillance **twice weekly** in fetuses with **end-diastolic velocities present** and **daily** in fetuses with **absent/reversed end-diastolic frequencies**

The optimal gestation to deliver the SGA fetus

- In the preterm SGA fetus with umbilical artery **AREDV** detected **prior to 32 weeks** of gestation, **delivery is recommended** when **DV Doppler becomes abnormal**, after completion of steroids.
- **Even when venous Doppler** is normal, delivery is recommended by **32 weeks** of gestation and should be considered between 30–32 weeks of gestation.
- . In the SGA fetus detected **after 32 weeks** of gestation with an abnormal umbilical artery Doppler, delivery **no later than 37 weeks** of gestation is recommended.
- In the SGA fetus detected **after 32 weeks of gestation** with **normal umbilical artery Doppler**, a senior obstetrician should be involved in determining the time and mode of birth

In the preterm SGA fetus, middle cerebral artery (MCA) Doppler has limited accuracy to predict acidemia and adverse outcome and should not be used to time delivery.

In the term SGA fetus with normal umbilical artery Doppler, an abnormal middle cerebral artery Doppler (PI < 5th centile) has moderate predictive value for acidosis at birth and should be used to time delivery.

Ductus venosus Doppler has moderate predictive value for acidemia and adverse outcome. Ductus venosus Doppler should be used for surveillance in the preterm SGA fetus with abnormal umbilical artery Doppler and used to time delivery.

How the SGA fetus should be delivered

- In the SGA fetus with umbilical artery AREDV delivery by caesarean section is recommended.
- In the SGA fetus with normal umbilical artery Doppler or with abnormal umbilical artery PI but end-diastolic velocities present, induction of labour can be offered but rates of emergency caesarean section are increased and continuous fetal heart rate monitoring is recommended from the onset of uterine contractions.

SFH

Single measurement <10th customised centile
and/or serial measurements indicative of FGR

High risk of SGA fetus/neonate
Based on history, biochemistry or uterine
artery Doppler

Fetal biometry

Single AC or EFW <10th customised centile
serial measurements indicative of FGR

UA Doppler

Normal

Repeat ultrasound (fortnightly)

AC and EFW¹² UA Doppler
MCA Doppler after 32 weeks

Delivery

Offer delivery by 37 weeks with the
involvement of a senior clinician
Recommend delivery by 37 weeks if
MCA Doppler PI <5th centile
Consider delivery >34 weeks if static growth
over 3 weeks
Recommend steroids if delivery is by CS
(as per RCO guidance)

PI or RI >2 SDs, EDV present

Repeat ultrasound

Weekly Twice weekly
AC and EFW¹² UA Doppler

Delivery

Recommend delivery by 37 weeks
Consider steroids if delivery by CS
Consider delivery >34 weeks if static growth
over 3 weeks
Recommend steroids if delivery is by CS
(as per RCOG guidance)

Refer for
fetal medicine
specialist
opinion

AREDV

Repeat ultrasound

Weekly Daily
AC and EFW¹² UA Doppler
DV Doppler
[cCTG]³

Delivery

Recommend delivery before 32 weeks after
steroids if:
– abnormal DV Doppler and/or cCTG
provided ≥24 weeks and EFW >500g
Recommend delivery by 32 weeks after steroids
Consider delivery at 30–32 weeks even when
DV Doppler is normal

An IUGR infant is at risk for

Hypothermia?

*decreased subcutaneous fat,
increased surface- volume
ratio, decreased heat
production*

Hypoglycemia?

Or

*decreased glycogen stores/
glycogenolysis/
gluconeogenesis*

Hypocalcemia?

*increased metabolic rate
deficient catecholamine release*

*Associated with perinatal stress,
asphyxia, prematurity*

Prognosis:

***The prognosis of FGR is highly dependent upon

1. The cause,
2. Severity and,
3. The gestation at delivery.

When FGR is related to a **congenital infection or chromosomal abnormality, subsequent development of the child will be determined by the **precise abnormality**.

Of babies with FGR secondary to **uteroplacental insufficiency, some babies will suffer morbidity or mortality as a result of prematurity for the survivors, the **long - term prognosis is good with low incidences of mental and physical disability** and most infants demonstrate ‘ **catch-up growth** ‘ after delivery when feeding is established.

****A link between FGR and the adult onset of hypertension and diabetes has been established.** It remains to be seen whether other associations will be found in the future.





THANKS